Lab Report
CSE 302 – Computer Organization & Architecture

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Task#1:

Convert by hand the number **`-123456789`** into its 32-bit single-precision binary representation, and then use the floating-point representation tool presented in Section 7.2 to verify your answer. Show your work for a full mark.

Sign = 1
Absolute value = 123456789
Binary = 111010110111100110100010101
Normalized = 1.11010110111100110100010101 × 2^26
Exponent (with bias 127) = 26 + 127 = 153 = 10011001
Fraction = 11010110111100110100010
32-bit float = 1 10011001 11010110111100110100010

Task#2: Convert by hand the floating-point number **`1 10010100 10011000001100000000000`** (shown in binary) into its corresponding decimal value, and then use the floating-point representation tool presented in Section 9.2 to verify your answer. Show your work for a full mark.

Sign = 1 → negative
Exponent = 10010100 = 148 → 148 - 127 = 21
Fraction = 1.10011000001100000000000
Decimal = - (1.100110000011 × 2^21) ≈ -3339699.84

Task#3: Trace the following program by hand to determine the values of registers **`$f0`** thru **`$f9`**. Notice that **`array1`** and **`array2`** have the same elements, but in a different order. Comment on the sums of **`array1`** and **`array2`** elements computed in registers **`$f4`** and **`$f9`**, respectively. Now use the MARS tool to trace the execution of the program and verify your results. What conclusion can be made from this exercise?

array1: 5.6e+20, -5.6e+20, 1.2
$f0 = 5.6e+20
$f1 = -5.6e+20
$f2 = 1.2
$f3 = $f0 + $f1 = 0
$f4 = $f3 + $f2 = 1.2
array2: 1.2, 5.6e+20, -5.6e+20
$f5 = 1.2
$f6 = 5.6e+20
$f7 = -5.6e+20
$f8 = $f5 + $f6 = 5.6e+20
$f9 = $f8 + $f7 = 0